Performance of the Models

	LR	HGB	RF	TabPFN	TRIFF
Training set ROC-AUC	0.7642 ± 0.0066	0.9996 ± 0.0001	1.0000 ± 0.0000	0.9635 ± 0.0020	0.8877 ± 0.0129
Training set AP	0.1610 ± 0.0081	0.9937 ± 0.0014	1.0000 ± 0.0000	0.6455 ± 0.0179	0.3296 ± 0.0390
Validation set ROC-AUC	0.7497 ± 0.0316	0.7975 ± 0.0245	0.8365 ± 0.0251	0.8952 ± 0.0165	0.8872 ± 0.0167
Validation set AP	0.1559 ± 0.0360	0.2228 ± 0.0253	0.2361 ± 0.0392	0.3567 ± 0.0511	0.3178 ± 0.0454
2025 test set ROC-AUC	0.7659 ± 0.0017	0.8418 ± 0.0210	0.8630 ± 0.0068	0.8687 ± 0.0073	0.8913 ± 0.0072
2025 test set AP	0.1605 ± 0.0017	0.2769 ± 0.0282	0.2464 ± 0.0175	0.3059 ± 0.0256	0.3778 ± 0.0486
Validation set - 2025 test set ROC-AUC	-0.0162 ± 0.0306	-0.0443 ± 0.0279	-0.0265 ± 0.0246	0.0266 ± 0.0166	-0.0041 ± 0.0176
Validation set - 2025 test set AP	-0.0046 ± 0.0361	-0.0540 ± 0.0367	-0.0103 ± 0.0329	0.0508 ± 0.0586	-0.0599 ± 0.0531

Mean ± sample SD across 15 repeated splits (not independent cohorts).
AP: sklearn average_precision_score; positive class: abnormal fertilization. Training/combined scores describe in-sample fitting, not held-out performance.
Difference: validation minus 2025 test, paired by split. LR: LinearRegression(fit_intercept=False, positive=True).